



연세대학교

Deep Learning Math · Lecture 1

Probability Foundations & Entropy

Why probability is the language of AI

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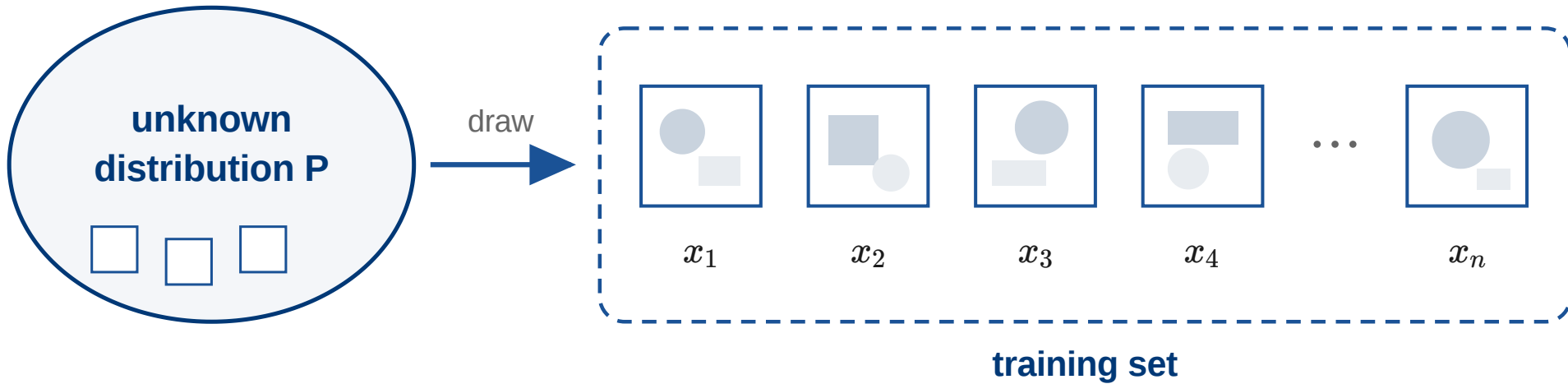
Jensen's inequality, $0 \leq H \leq \log M$

01

Why Probability?

The language of AI

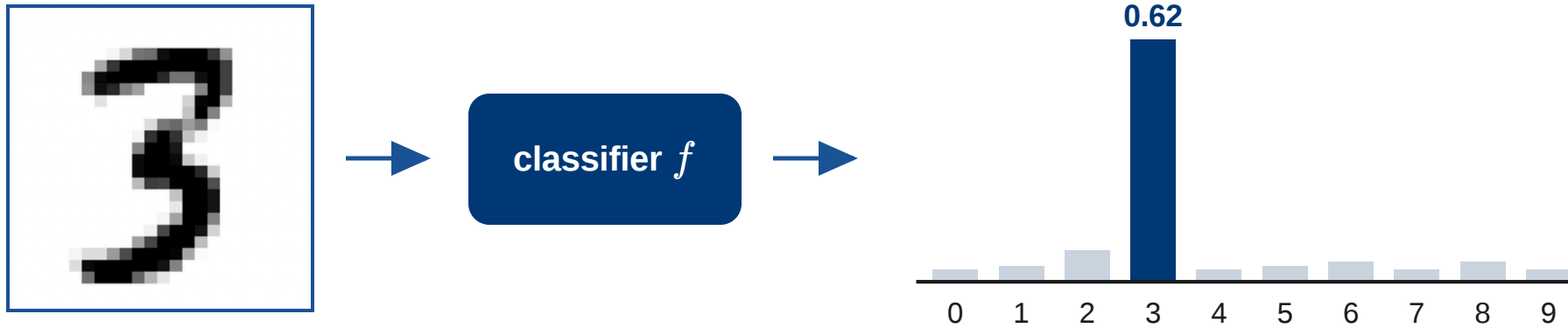
Collecting Data = Sampling



$$x_1, x_2, \dots, x_n \stackrel{\text{i.i.d.}}{\sim} P$$

P is never revealed — only its samples are.

Models Learn Distributions



- Classifier output = **pmf over labels**
- Generative model = distribution over images
- RL policy = distribution over actions

02

Probability Review

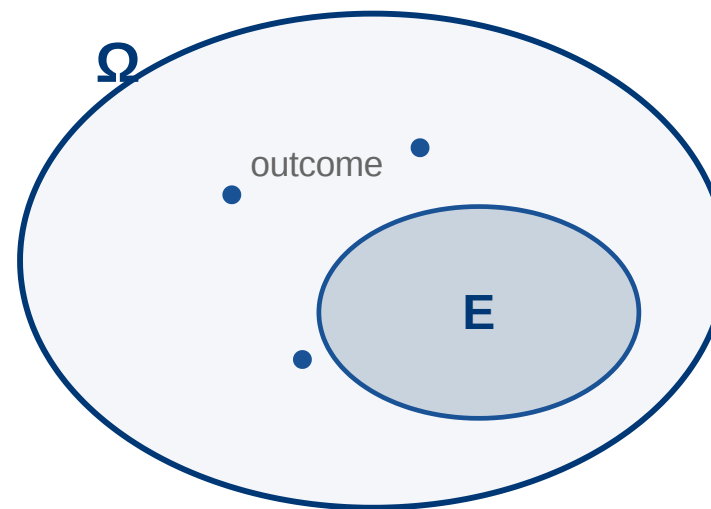
Sample space, events, probability measure

Sample Space and Events

Sample space Ω : set of all possible outcomes

Event E : a (measurable) subset of Ω

Event collection $\mathcal{F}$: set of events



Probability Measure

Assigns each event a number in $[0, 1]$:

$$P : \mathcal{F} \rightarrow [0, 1]$$

- $P(\emptyset) = 0, P(\Omega) = 1$
- Larger event, larger probability

Triple $(\Omega, \mathcal{F}, P)$: the full probabilistic model.

Example — Fair Coin

Sample space $\Omega = \{H, T\}$, events:

$$\mathcal{F} = \{ \emptyset, \{H\}, \{T\}, \{H, T\} \}$$

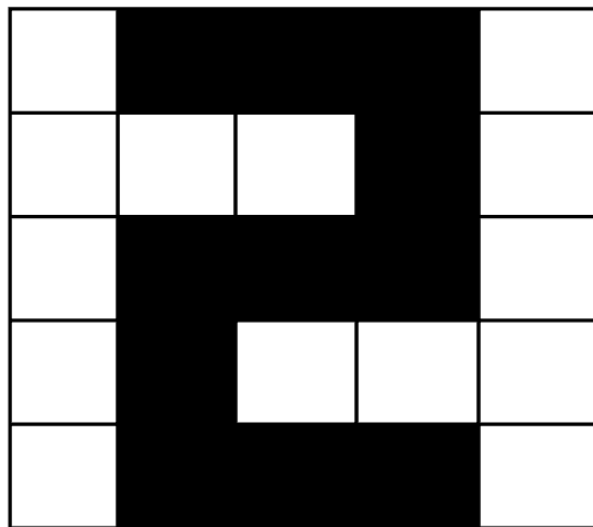
Probability measure:

$$P(\emptyset) = 0, \quad P(\{H\}) = P(\{T\}) = \frac{1}{2}, \quad P(\{H, T\}) = 1$$

Every subset gets a probability: four events total.

Example — 5×5 Binary Image

An image = a 25-dimensional binary vector:

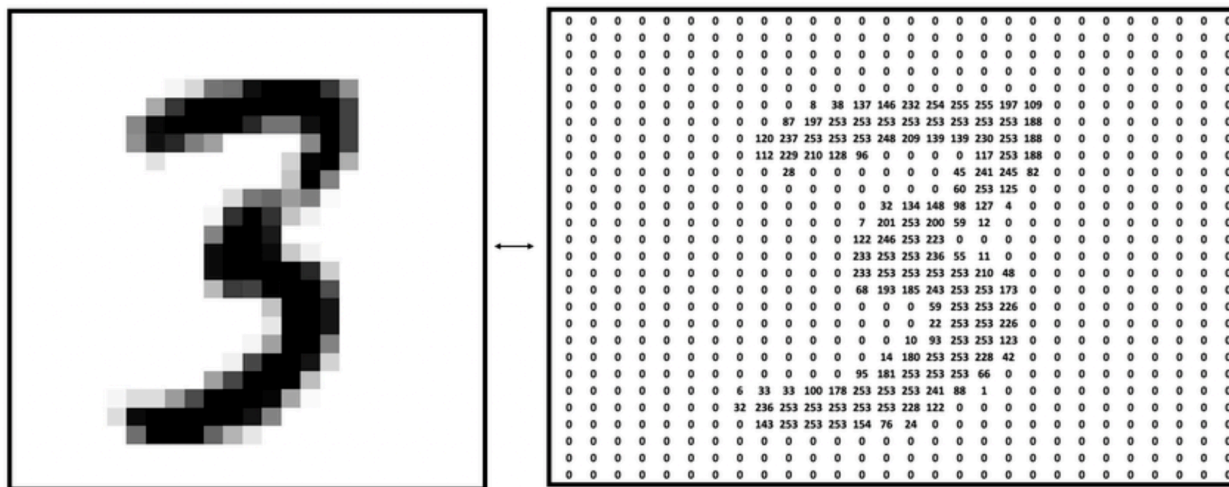


0	1	1	1	0
0	0	0	1	0
0	1	1	1	0
0	1	0	0	0
0	1	1	1	0

$$\Omega = \{0, 1\}^{5 \times 5}, \quad |\Omega| = 2^{25}, \quad |\mathcal{F}| = 2^{2^{25}}$$

Example — MNIST

Grayscale 28×28, pixel intensity 0 to 255:

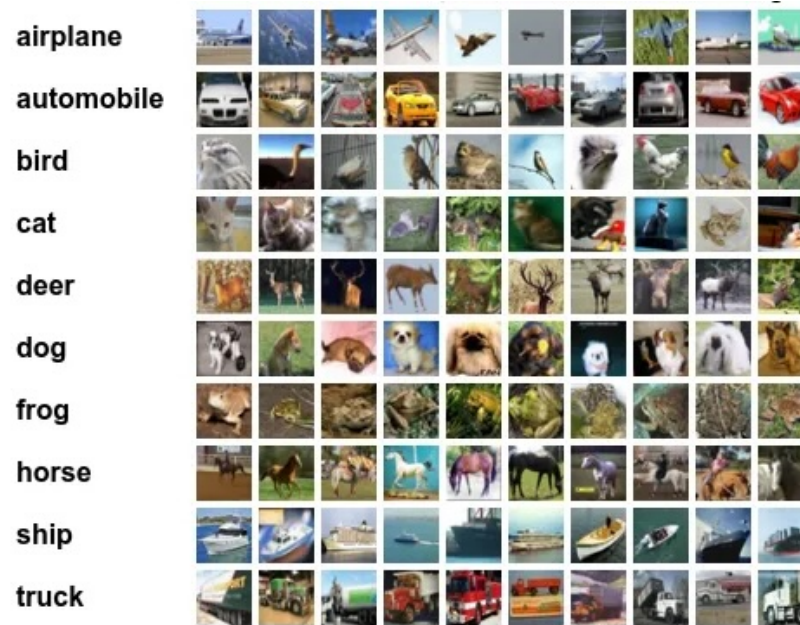


$$\Omega = \{0, 1, \dots, 255\}^{28 \times 28}$$

The probability on Ω is **not uniform** (of course)

Example — CIFAR-10

RGB color 32×32, three channels:

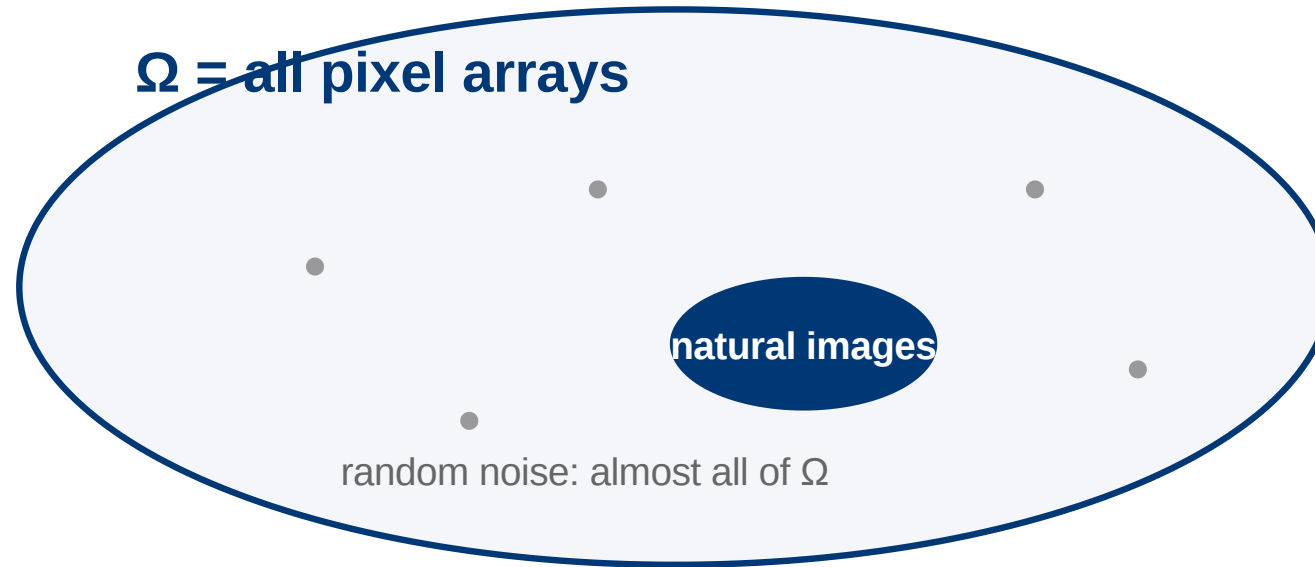


$$\Omega = \{0, 1, \dots, 255\}^{3 \times 32 \times 32}$$

Sample Spaces of Real Data

DATASET	Ω	$ \Omega $
Fair coin	$\{H, T\}$	2
5×5 binary	$\{0, 1\}^{25}$	$2^{25} \approx 3.4 \times 10^7$
MNIST	$\{0, \dots, 255\}^{28 \times 28}$	$2^{6272} \approx 10^{1888}$
CIFAR-10	$\{0, \dots, 255\}^{3 \times 32 \times 32}$	$2^{24576} \approx 10^{7398}$
ImageNet	$\{0, \dots, 255\}^{3 \times 224 \times 224}$	$2^{1204224} \approx 10^{362508}$

Astronomical, and Non-Uniform



- Atoms in the universe: roughly 10^{80}
- Almost every element of Ω : pure noise

Data distributions put **huge mass on tiny regions**: that structure is what models learn.

03

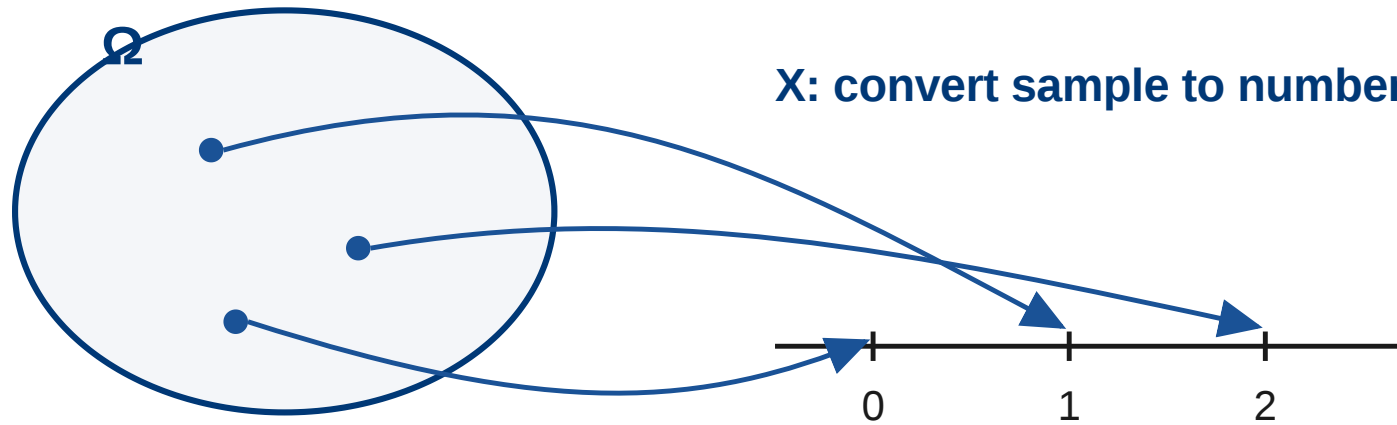
Random Variables

From outcomes to numbers

Random Variable

A map from sample space to real numbers:

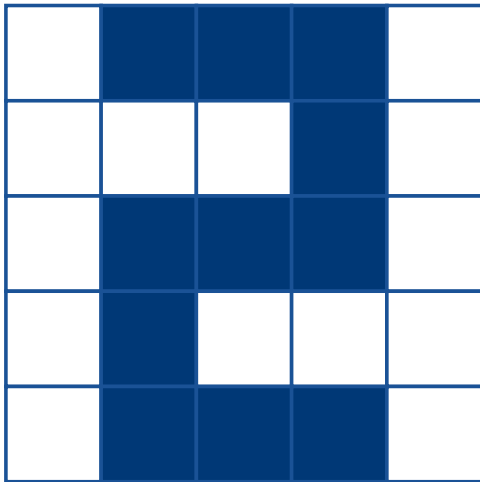
$$X : \Omega \rightarrow \mathbb{R}$$



Uppercase X : the random variable. Lowercase x : a realization.

Example — Count of Ones

On $\Omega = \{0, 1\}^{5 \times 5}$: let X = number of 1's



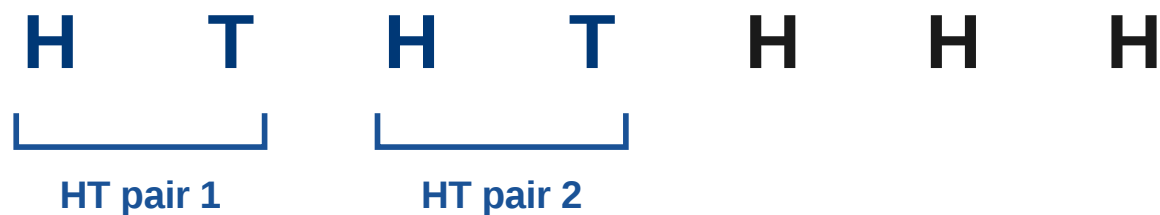
This outcome ω : the digit "2"

$$X(\omega) = 11$$

Eleven blue pixels: one number per image.

Example — Seven Coin Tosses

On $\Omega = \{H, T\}^7$: let X = number of heads followed by a tail



$$X(HTHTHHH) = 2$$

Probability Mass Function

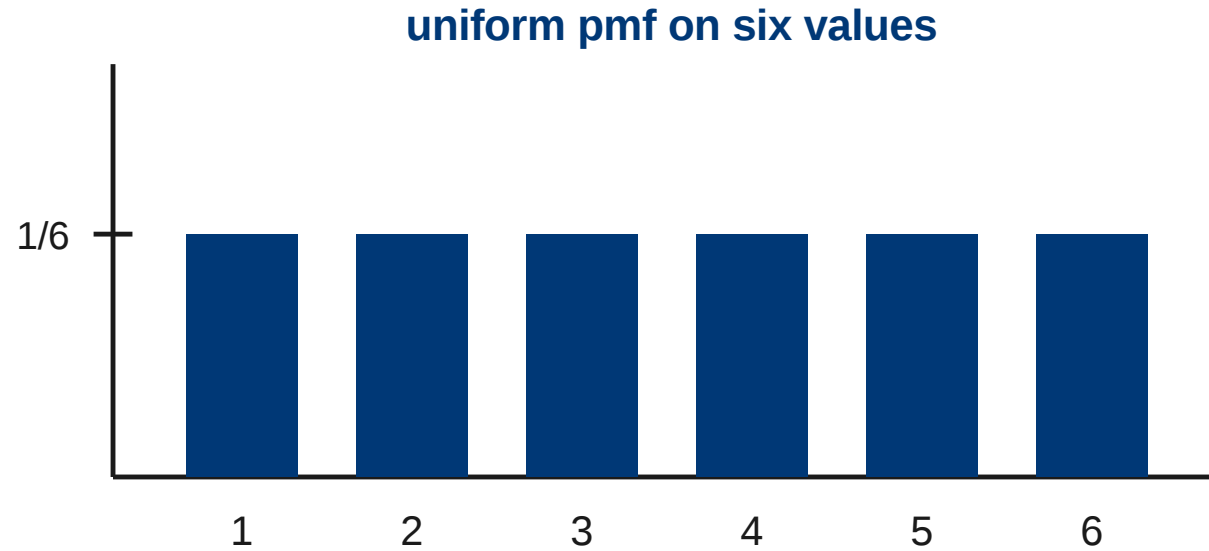
The pmf of a random variable X :

$$p_X(x) = \Pr(X = x)$$

- $\mathcal{X}$: set of possible values of X
- Uppercase X random, lowercase x realization

The pmf **completely characterizes** the random variable.

Example — pmf of a Fair Die



$$p_X(x) = \frac{1}{6}, \quad x \in \mathcal{X} = \{1, 2, 3, 4, 5, 6\}$$

Expectation

Probability-weighted average of the values:

Definition (expectation).

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x \cdot p_X(x)$$

Fair die:

$$\mathbb{E}[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5$$

Expectation of a Function

Exercise. For any $f : \mathcal{X} \rightarrow \mathbb{R}$, show

$$\mathbb{E}[f(X)] = \sum_{x \in \mathcal{X}} f(x) \cdot p_X(x)$$

Hint: set a new random variable $Y = f(X)$

We use this **constantly** today: entropy is $\mathbb{E}[f(X)]$ for one special f .

04

Entropy

Measuring uncertainty

How Surprising Is an Outcome?

Indicator X : $X = 1$ serious earthquake, $X = 0$ otherwise

$X = 0$ (typical day)

$$p_X(0) \approx 1$$

No surprise at all

$X = 1$ (earthquake)

$$p_X(1) \text{ small}$$

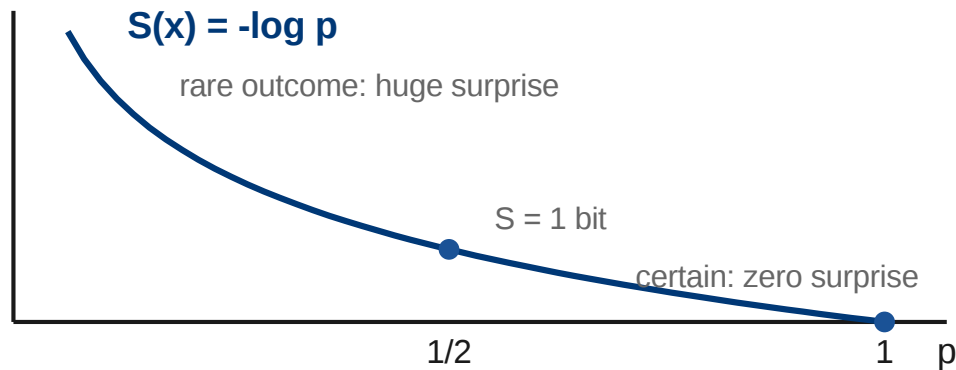
Very surprising

Rare event = surprising. Surprise should grow as $p_X(x)$ shrinks.

Surprisal

Definition (surprisal).

$$S(x) = \log \frac{1}{p_X(x)} = -\log p_X(x)$$



Why the Logarithm?

- Certain outcome: $p = 1 \Rightarrow S = 0$
- Rarer outcome, larger surprise: S decreasing in p
- Independent surprises **add**:

$$\log \frac{1}{p \cdot q} = \log \frac{1}{p} + \log \frac{1}{q}$$

Deeper axiomatic justification exists (information theory); other measures too, e.g. Renyi entropy.

Entropy — Definition

Expected surprisal of X with $\mathcal{X} = \{0, 1, \dots, M - 1\}$:

$$\begin{aligned} H(X) &= \mathbb{E}[S(X)] \\ &= \mathbb{E}[-\log p_X(X)] \\ &= \sum_x p_X(x) \log \frac{1}{p_X(x)} \end{aligned}$$

Conventions: $0 \cdot \log 0 = 0$; $\log$ is base 2; unit is **bits**

Entropy Reads Two Ways

Uncertainty

How unpredictable is X before observing?

Information

How much do we learn after observing?

Same number: **resolving uncertainty = gaining information.**

Worked Example — Fair Coin

$$p_X(H) = p_X(T) = 1/2:$$

$$H(X) = \frac{1}{2} \log 2 + \frac{1}{2} \log 2 = 1 \text{ bit}$$

One fair coin toss = **exactly one bit** of uncertainty.

Worked Example — Biased Coin

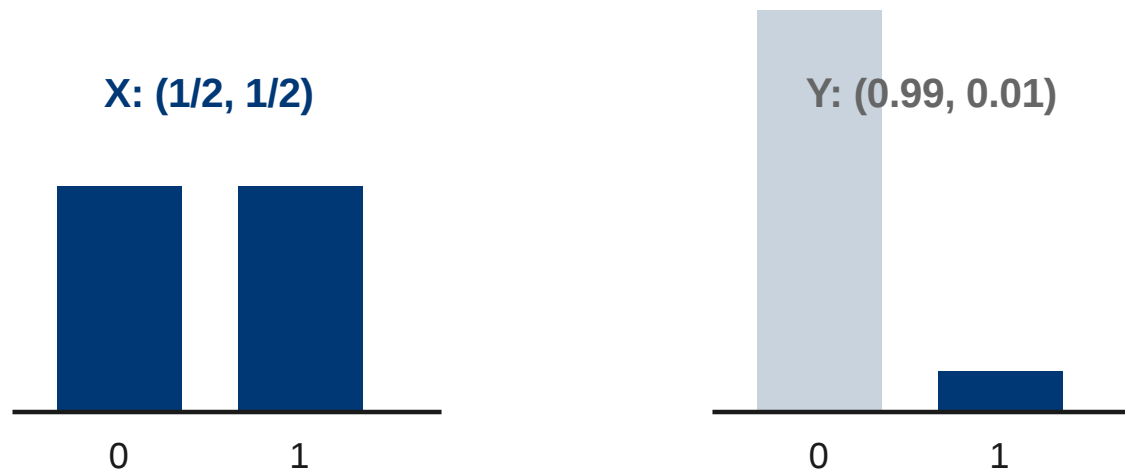
$p_X(0) = 0.01, p_X(1) = 0.99$:

$$\begin{aligned} H(X) &= 0.01 \log \frac{1}{0.01} + 0.99 \log \frac{1}{0.99} \\ &\approx 0.066 + 0.014 \approx 0.081 \text{ bits} \end{aligned}$$

Rare outcome: surprisal $\log 100 \approx 6.64$ bits, but weight only 0.01

Which Is More Random?

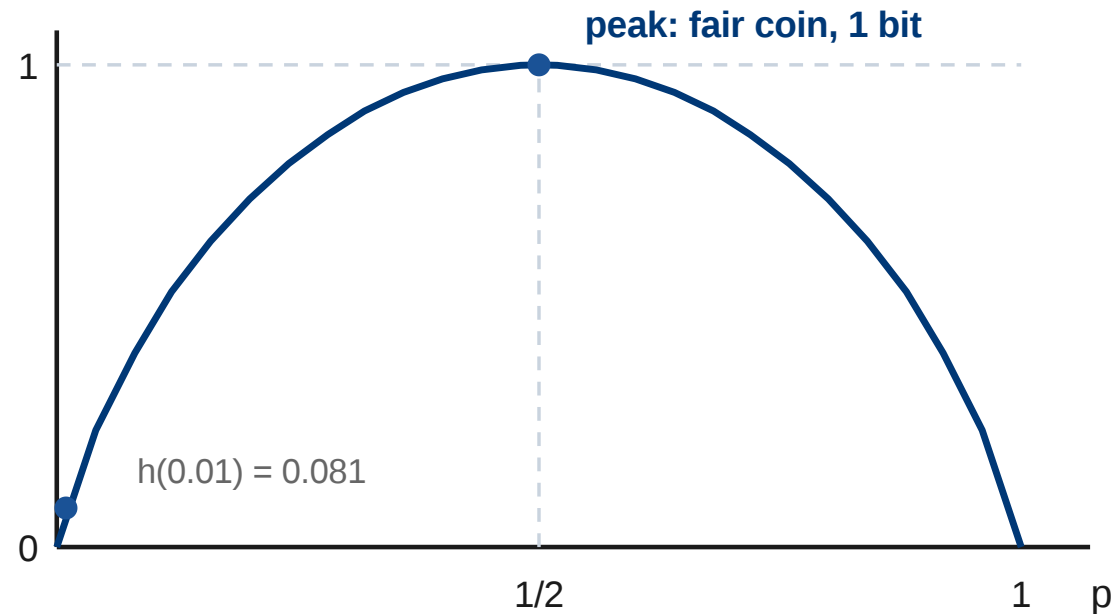
Both binary, $\mathcal{X} = \{0, 1\}$:



Entropy agrees with intuition: X far more uncertain than Y

Binary Entropy Function

Bernoulli(p): $h_2(p) = -p \log p - (1 - p) \log(1 - p)$



Symmetric: $h_2(p) = h_2(1 - p)$; zero at both ends.

Worked Example — Fair Die

Uniform on six faces, $p_X(x) = 1/6$:

$$H(X) = \sum_{x=1}^6 \frac{1}{6} \log 6 = \log 6 \approx 2.585 \text{ bits}$$

Uniform pmf: entropy is just log of the alphabet size. Keep this in mind.

Worked Example — Three Outcomes

$p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$: surprisals 1, 2, 2 bits

$$H(X) = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{4} \cdot 2 = 1.5 \text{ bits}$$

Between coin (1 bit) and uniform-on-3 ($\log 3 \approx 1.585$)

Guessing Game

Alice picks a number 0 to 7; Bob asks yes/no questions



Q1: is it in $\{0, 1, 2, 3\}$? **Q2:** is it in $\{3, 6\}$?

Which Question Is More Informative?

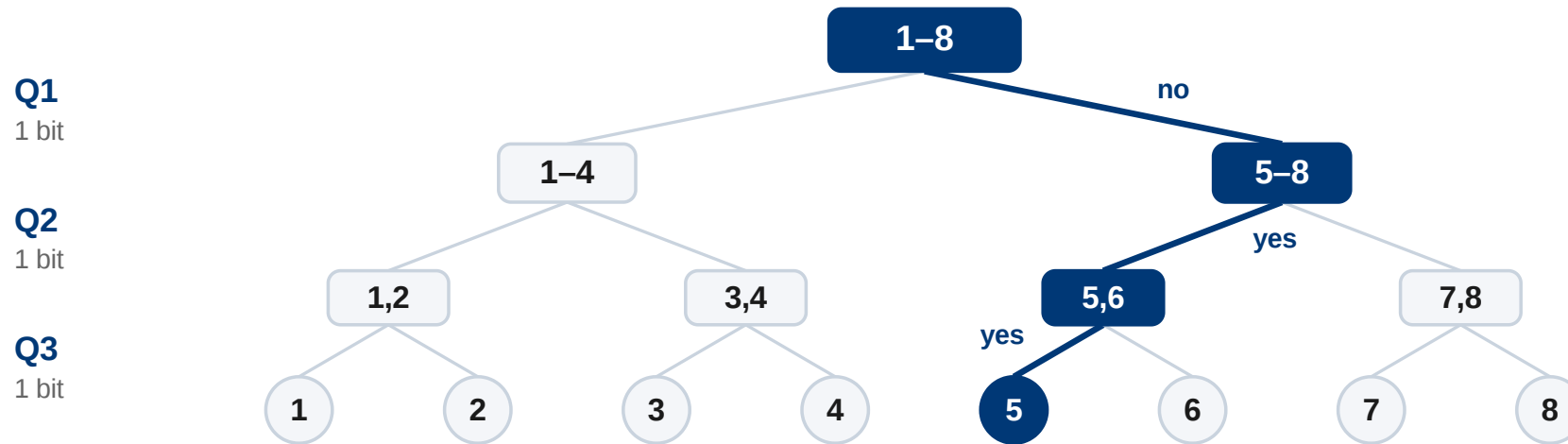
Entropy of the yes/no **answer**:

$$H(\text{answer to Q1}) = h_2\left(\frac{1}{2}\right) = 1 \text{ bit}$$

$$H(\text{answer to Q2}) = h_2\left(\frac{1}{4}\right) \approx 0.811 \text{ bits}$$

Q1 wins: higher-entropy answer = more informative question.

Twenty Questions — One Bit Each

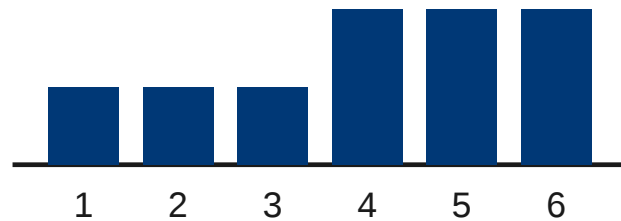


$$3 \text{ balanced questions} = \log 8 = H(X)$$

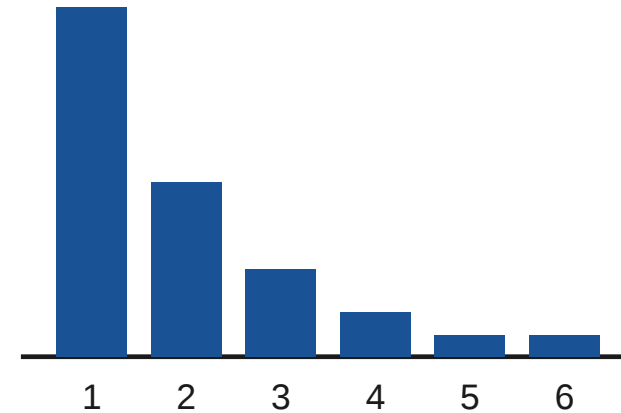
Each answer halves the candidates: exactly one bit.

Two Dice — Which More Uncertain?

D1: $(1/9, 1/9, 1/9, 2/9, 2/9, 2/9)$



D2: $(1/2, 1/4, 1/8, 1/16, 1/32, 1/32)$



Same six faces, different pmfs. Guess first, then compute.

Two Dice — Computing H

$$H(D_1) = 3 \cdot \frac{1}{9} \log 9 + 3 \cdot \frac{2}{9} \log \frac{9}{2} \approx 2.503 \text{ bits}$$

$$H(D_2) = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{16} \cdot 4 + \frac{2}{32} \cdot 5 = \frac{31}{16} \approx 1.938$$

D_1 is more uncertain: closer to uniform

Both below $\log 6 \approx 2.585$: a hint of things to come.

AI Example — Classifier Soft Labels

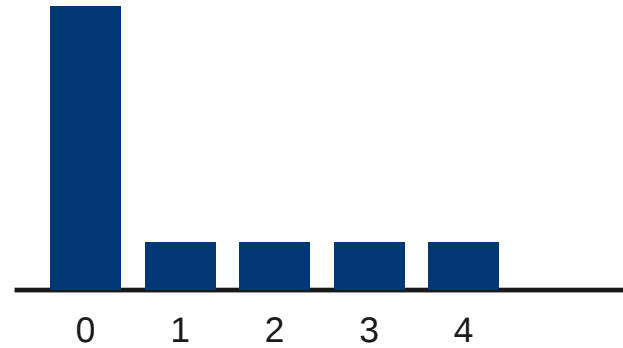
Digit classifier f (labels 0 to 4) outputs a **pmf**:

$$f(x) = (0.4, 0.4, 0.2, 0, 0)$$

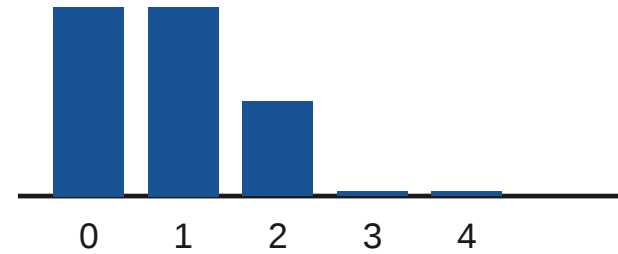
- Reads: digit "0" w.p. 0.4, digit "2" w.p. 0.2
- Soft guessing, like a weather forecast

Which Sample Is Harder?

$$f(x_1) = (0.6, 0.1, 0.1, 0.1, 0.1)$$



$$f(x_2) = (0.4, 0.4, 0.2, 0, 0)$$



$$H(f(x_1)) \approx 1.771 \text{ bits}, \quad H(f(x_2)) \approx 1.522 \text{ bits}$$

Entropy ranks x_1 as more uncertain: mass spread over all five labels.

AI Example — Data Valuation

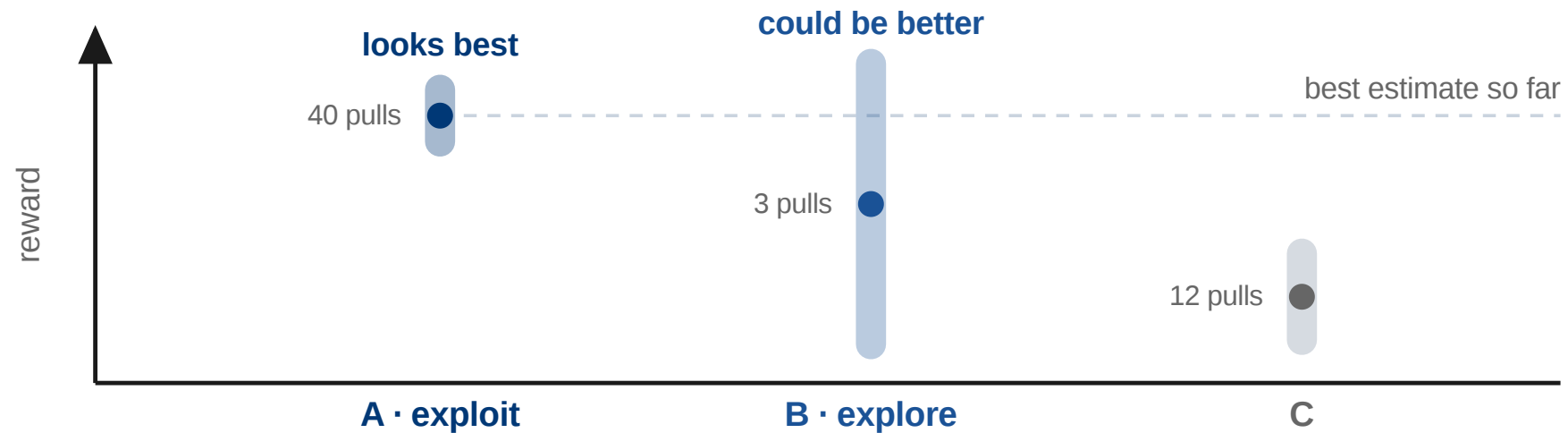
Which sample is most **helpful for training**?



- Challenging samples teach the model most
- High-entropy prediction = model still confused there

AI Example — Explore vs Exploit

Each round: pick one strategy, see one noisy reward



Never trying B means never learning B is better.

Entropy Measures Explorativeness

Go strategies A to E; current belief: A is best

$$S_1 = (0.6, 0.1, 0.1, 0.1, 0.1)$$

$$S_2 = (0.4, 0.4, 0.2, 0.0, 0.0)$$

- Exploit A, but reserve probability for the rest
- $H(S_1) \approx 1.771 > H(S_2) \approx 1.522$: S_1 more explorative

Entropy = a measure of **explorativeness** of a policy.

Exercise — Geometric Distribution

Let X be geometric with $p = 1/2$:

$$p_X(x) = \frac{1}{2^x}, \quad x \geq 1$$

Exercise. What is $H(X)$?

Hint: surprisal of outcome x is exactly x bits, so $H(X) = \mathbb{E}[X]$

05

Properties of Entropy

Jensen's inequality and $0 \leq H \leq \log M$

Theorem — Non-Negativity

Theorem 1 (non-negativity).

Entropy is non-negative:

$$H(X) \geq 0$$

with equality if and only if X is deterministic.

A sanity floor: no distribution has negative uncertainty.

Proof — Non-Negativity

Every probability satisfies $0 \leq p_X(x) \leq 1$:

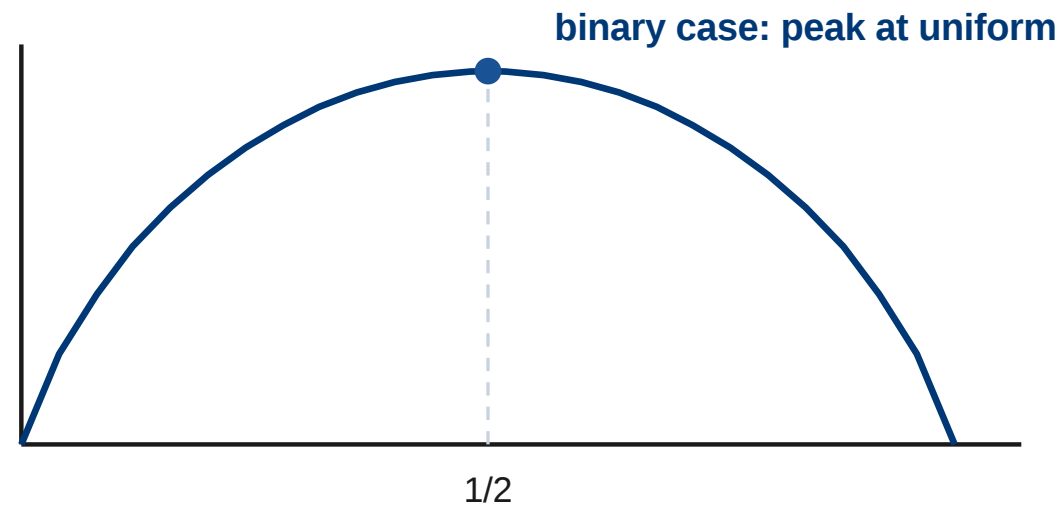
$$H(X) = \mathbb{E} \left[\log \frac{1}{p_X(X)} \right] \stackrel{(p_X \leq 1)}{\geq} \mathbb{E}[\log 1] = 0 \quad \blacksquare$$

Equality: needs $p_X(x) \log p_X(x) = 0$ for all x

So every $p_X(x)$ is 0 or 1: X deterministic

Question — Which pmf Maximizes H ?

Floor is settled. What about the **ceiling**?



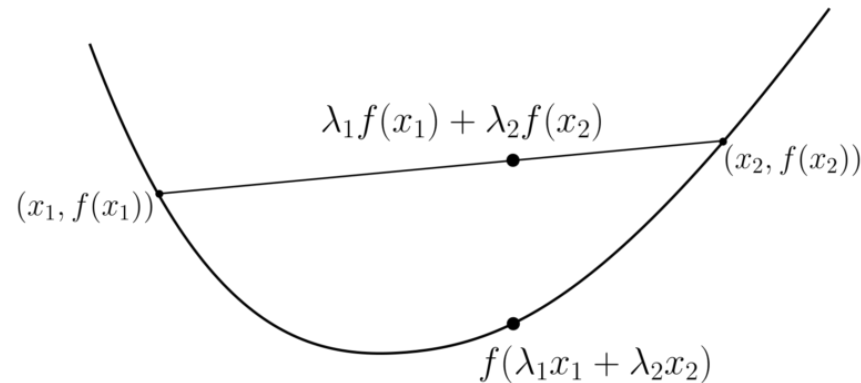
Guess: **uniform** maximizes entropy. To prove it we need a tool.

Technique Review — Convex Functions

Definition (convex).

f is convex if for all x, y and $0 \leq \lambda \leq 1$:

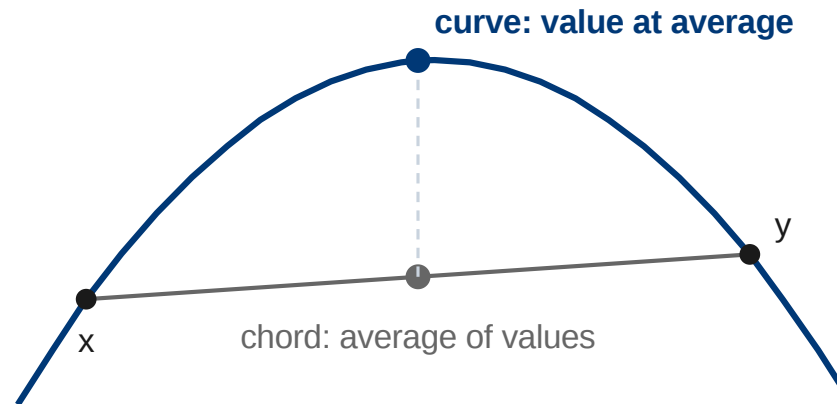
$$\lambda f(x) + (1 - \lambda)f(y) \geq f(\lambda x + (1 - \lambda)y)$$



Technique Review — Concave Functions

Flip the inequality: f concave if

$$\lambda f(x) + (1 - \lambda)f(y) \leq f(\lambda x + (1 - \lambda)y)$$



Chord under the curve. Key example: $\log$ is concave.

Convex or Concave?

$$x^2$$

Convex: chord above

$$\log x$$

Concave: chord below

$$h_2(p)$$

Concave: the arch
we plotted

Exercise. Show the binary entropy $h_2(p)$ is concave

Hint: second derivative.

Jensen's Inequality — Statement

Theorem 2 (Jensen's inequality).

Let f be concave. Then

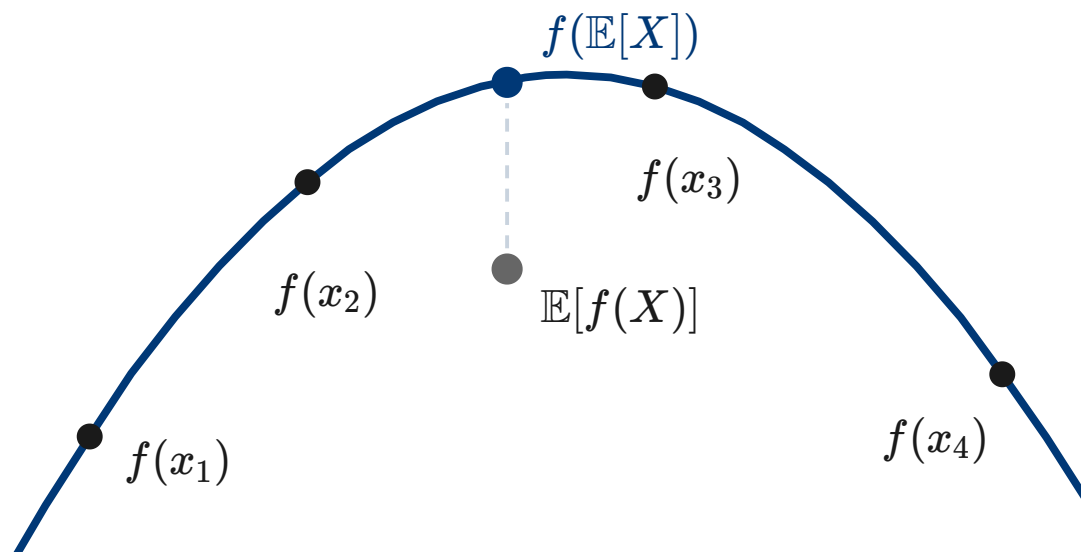
$$\mathbb{E}[f(X)] \leq f(\mathbb{E}[X])$$

For strictly concave f : equality iff X is constant.

For convex f the inequality flips.

Jensen — Reading the Statement

Concavity compares **two** points; Jensen compares **any pmf**:



Weighted average of f values never beats f at the weighted average input.

Jensen — Proof Overview

Target, for pmf $(p_1, \dots, p_M)$ on points $(x_1, \dots, x_M)$:

$$\sum_{i=1}^M p_i f(x_i) \leq f\left(\sum_{i=1}^M p_i x_i\right)$$

Induction on the support size M :

1. Base $M = 2$: exactly the definition of concavity
2. Step: peel off one atom, renormalize the rest

Jensen Proof — Base Case

$M = 2$: pmf (p_1, p_2) with $p_2 = 1 - p_1$

$$p_1 f(x_1) + (1 - p_1) f(x_2) \stackrel{(\text{concavity, } \lambda=p_1)}{\leq} f(p_1 x_1 + (1 - p_1) x_2)$$

Left side is $\mathbb{E}[f(X)]$, right side is $f(\mathbb{E}[X])$

The definition of concavity, wearing expectation notation.

Jensen Proof — Peel One Atom

Assume true for $M - 1$ points. Split off atom M (take $p_M < 1$):

$$\sum_{i=1}^M p_i f(x_i) = p_M f(x_M) + (1 - p_M) \sum_{i=1}^{M-1} q_i f(x_i), \quad q_i = \frac{p_i}{1 - p_M}$$

The q_i sum to one: a **valid pmf on $M - 1$ points**

Renormalizing is the whole trick: the rest is bookkeeping.

Jensen Proof — Induction Step

Recap: peeled form $p_M f(x_M) + (1 - p_M) \sum_i q_i f(x_i)$

$$\begin{aligned} p_M f(x_M) + (1 - p_M) \sum_{i=1}^{M-1} q_i f(x_i) &\stackrel{(\text{IH on } q)}{\leq} p_M f(x_M) + (1 - p_M) f\left(\sum_{i=1}^{M-1} q_i x_i\right) \\ &\stackrel{(\text{concavity})}{\leq} f\left(p_M x_M + (1 - p_M) \sum_{i=1}^{M-1} q_i x_i\right) \\ &= f\left(\sum_{i=1}^M p_i x_i\right) = f(\mathbb{E}[X]) \quad \blacksquare \end{aligned}$$

Jensen — Proof Summary

$$\mathbb{E}[f(X)] \stackrel{(\text{peel})}{=} p_M f(x_M) + (1 - p_M) \mathbb{E}_q[f] \stackrel{(\text{IH})}{\leq} \dots \stackrel{(\text{2-pt def.})}{\leq} f(\mathbb{E}[X])$$

- **Key trick:** renormalize the rest into a smaller pmf
- Two-point concavity does all the lifting
- Equality (strict f): all mass at one point

Jensen — Why We Care

- **Today:** maximum entropy, $H(X) \leq \log M$
- **Next lecture:** KL divergence is non-negative
- **Everywhere in ML:** log-likelihood lower bounds (ELBO)

One-line recipe: expectation trapped under a log? **Push it inside.**

Theorem — Maximum Entropy

Theorem 3 (maximum entropy).

For a random variable $X \in \mathcal{X}$ with $|\mathcal{X}| = M$:

$$H(X) \leq \log |\mathcal{X}| = \log M$$

with equality if and only if X is uniform.

The ceiling we guessed: uniform is the most uncertain pmf.

Max Entropy — Proof Overview

Three moves to $H(X) \leq \log M$, with $Z = 1/p_X(X)$:

1 Rewrite $H(X) = \mathbb{E}[\log Z]$

2 Jensen $\mathbb{E}[\log Z] \leq \log \mathbb{E}[Z]$ — log is concave

3 Cancel $\mathbb{E}[Z] = \sum_x p_X(x) \cdot \frac{1}{p_X(x)} = M$

Step 3 is the punchline: the pmf cancels itself.

Proof — Apply Jensen

The random quantity inside is $\frac{1}{p_X(X)}$; $\log$ is concave:

$$H(X) = \mathbb{E} \left[\log \frac{1}{p_X(X)} \right] \stackrel{(\text{Jensen})}{\leq} \log \mathbb{E} \left[\frac{1}{p_X(X)} \right]$$

One unknown left: the expectation on the right

Proof — The Cancellation

Expand the expectation over x with weights $p_X(x)$:

$$\mathbb{E}\left[\frac{1}{p_X(X)}\right] \stackrel{(\mathbb{E}[f(X)])}{=} \sum_{x=1}^M p_X(x) \cdot \frac{1}{p_X(x)} = \sum_{x=1}^M 1 = M$$

$$\therefore H(X) \leq \log M \quad \blacksquare$$

Weight $p_X(x)$ meets value $1/p_X(x)$: each term is exactly 1.

Proof — Equality Condition

Jensen (strictly concave $\log$) is tight iff the inside is **constant**:

$$\frac{1}{p_X(x)} = c \text{ for all } x \implies p_X(x) = \frac{1}{M}$$

Constant pmf summing to one: the **uniform** distribution

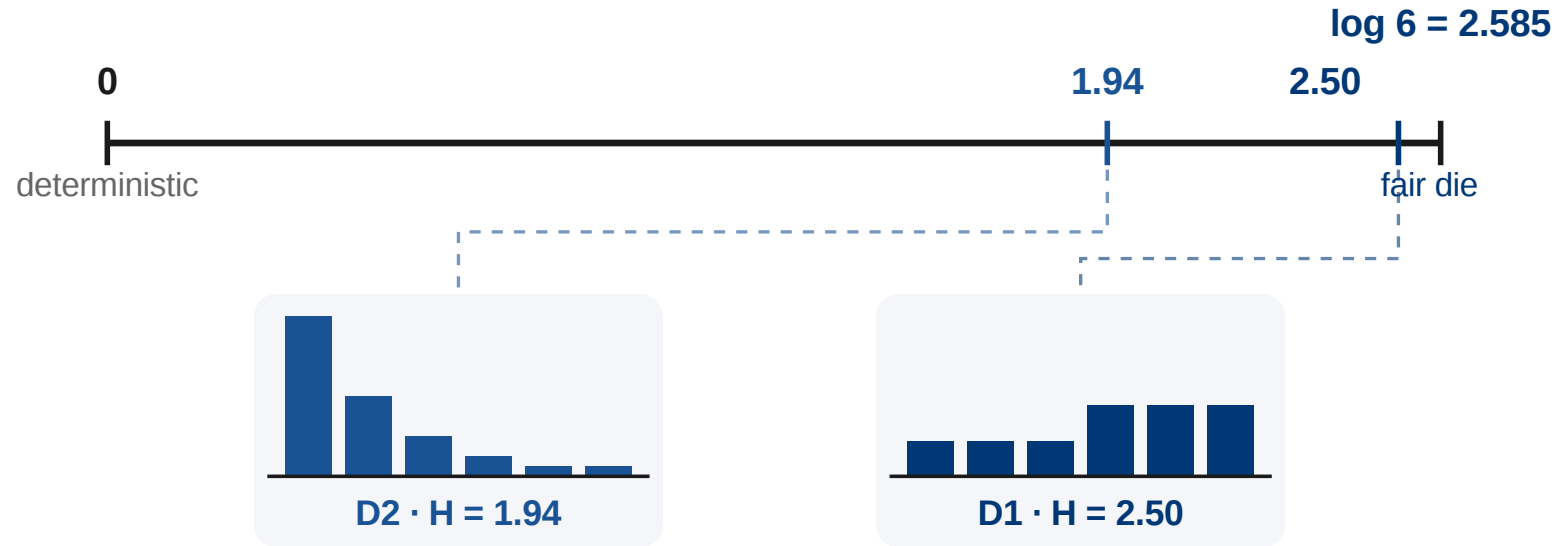
Matches the fair die: $H = \log 6$ exactly.

Max Entropy — Proof Summary

$$H(X) = \mathbb{E} \left[\log \frac{1}{p_X(X)} \right] \stackrel{(\text{Jensen})}{\leq} \log \mathbb{E} \left[\frac{1}{p_X(X)} \right] \stackrel{(\text{cancel})}{=} \log M$$

- **Key trick:** $1/p_X$ inside its own expectation cancels
- Jensen direction set by concavity of $\log$
- Tight iff p_X uniform

Implication — The Entropy Scale



$$0 \leq H(X) \leq \log M$$

Implication — Bits to Describe X

- $\log M$ = bits to index M items
- Guessing game: 8 numbers, $\log 8 = 3$ questions
- Uniform Alice is the **hardest opponent**

Uniform = maximal ignorance: no structure to exploit.

Sanity Check — All Examples

DISTRIBUTION	H (BITS)	$\text{LOG } M$
Fair coin	1	$\log 2 = 1$ (tight!)
Biased coin (0.01, 0.99)	0.081	1
Dice D_1, D_2	2.503, 1.938	$\log 6 \approx 2.585$
Soft labels $f(x_1), f(x_2)$	1.771, 1.522	$\log 5 \approx 2.322$

Tight exactly when uniform, as the theorem promised.

Recap — Today's Chain

$$(\Omega, \mathcal{F}, P) \xrightarrow{X: \Omega \rightarrow \mathbb{R}} p_X(x) \xrightarrow{\text{surprisal}} S(x) = \log \frac{1}{p_X(x)}$$
$$H(X) = \mathbb{E}[S(X)], \quad 0 \stackrel{(\text{Thm 1})}{\leq} H(X) \stackrel{(\text{Jensen})}{\leq} \log M$$

- Datasets = sample spaces; models = distributions
- Entropy: expected surprisal, in bits
- Extremes: deterministic vs uniform

Next — The Wrong Distribution

Surprisal used the **true** pmf p_X . But a model only has a guess Q :

$$\mathbb{E} \left[\log \frac{1}{Q(X)} \right] \stackrel{?}{\geq} H(X)$$

- How much does the mismatch cost?
- Same Jensen trick, one more twist

Next lecture: **cross-entropy and KL divergence** — the loss function of deep learning.

$$H(X) \leq \log M$$

Questions?